

(a)

Identity only



$$\mathcal{A}'_X = M : [M, A] = 0$$

commutant dimension = 4
unresolved multiplicity

full-matrix mixing obstructed

(b)

Diagonal generator



$$\mathcal{A}'_X = M : [M, A] = 0$$

commutant dimension = 2
invariant blocks

full-matrix mixing obstructed

(c)

Pauli pair



$$\mathcal{A}'_X = M : [M, A] = 0$$

commutant dimension = 1
irreducible response

full matrix algebra allowed

Scalar commutant is necessary, not sufficient: irreducibility does not imply Gaussianity, ETH, or thermalization.